2.b.	**Give** any three built-in functions for string handling and character handling each, along with their syntax and examples.		06	L2	CO2
2.c.	**How** a single dimension and two-dimension arrays are declared and initialized in C.		05	L2	CO2
3.a.	**Differentiate** call-by-value and call-by-reference with suitable C code snippets.		06	L2	CO3
3.b.	**Explain** recursion with a programming example.		05	L2	CO3
3.c.	**Develop** a C program to swap two numbers using pointers.		05	L3	CO3
4.a.	**Develop** a C program using nested structures to store student details (Name, Roll No, DOB, Marks in 3 subjects) and compute total marks.		08	L3	CO4
4.b.	**Differentiate** between structures and unions with examples.		08	L2	CO4
5.a.	**Outline** the key differences between malloc(), calloc(), realloc(), and free() in C with syntax and examples		08	L2	CO5
5.b.	**Develop** a C program to create a file, validate file creation and write 26 characters (A–Z) into it.		08	L3	CO5

DAYANANDA SAGAR UNIVERSITY

SCHOOL OF ENGINEERING

USN No: ENG3CS0453

II Semester B.Tech Mid Semester Examination-AY 2025-26

Electronics and Communication Engineering

Course Title: Introduction to Electrical and Electronics Engineering **Course Code:** 25EN1109

Duration: 75 Mins.

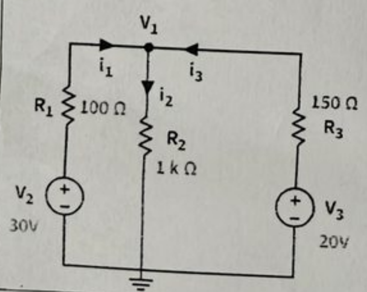
Date: 14/03/2026

Time: 09:00 – 10:15 AM

Max Marks: 40

Instructions to Candidates:

- Answer all the four Questions
- Each question carries 10 marks

Q.No.	QUESTION	Marks	BT Level	CO's
1(a)	Calculate the electrical energy in kWh consumed in a month, in a house using 2 bulbs of 100 W each and 2 fans of 60 W each, if the bulbs and fans are used for an average of 10 hours each day. Determine the electricity bill if the charges are Rs. 2/unit.	05	L3	CO1
1(b)	Explain the concept of active and passive circuit elements and list examples for each.	05	L2	CO1
2(a)	Explain the series and parallel resistor networks with formulas for equivalent resistance.	06	L2	CO1
2(b)	Determine the currents i_1, i_2 and i_3 for the given circuit using KCL.  	04	L3	CO1
3(a)	Explain the circuit configuration and operation of a half-wave rectifier with a reservoir capacitor using suitable waveforms.	06	L2	CO3